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Survey paper

A review of visual SLAM methods for autonomous driving vehicles

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ABSTRACT

Autonomous driving vehicles require both a precise localization and mapping solution in different driving environment. In this context, Simultaneous Localization and Mapping (SLAM) technology is a well-study settlement. Light Detection and Ranging (LIDAR) and camera sensors are commonly used for localization and perception. However, through ten or twenty years of evolution, the LIDAR-SLAM method does not seem to have changed much. Compared with the LIDAR based schemes, the visual SLAM has a strong scene recognition ability with the advantages of low cost and easy installation. Indeed, people are trying to replace LIDAR sensors with camera only, or integrating other sensors on the basis of camera in the field of autonomous driving. Based on the current research situation of visual SLAM, this review covers the visual SLAM technologies. In particular, we firstly illustrated the typical structure of visual SLAM. Secondly, the state-of-the-art studies of visual and visual-based (i.e. visual-inertial, visual-LIDAR, visual-LIDAR-IMU) SLAM are completely reviewed, as well the positioning accuracy of our previous work are compared with the well-known frameworks on the public datasets. Finally, the key issues and the future development trend of visual SLAM technologies for autonomous driving vehicles applications are discussed.

1. Introduction

With the development of robotics and artificial intelligence (AI) technology, the autonomous driving vehicles (cars) has been an attractive subject in industry and academia (Badue et al., 2021). In order to navigate safely, it needs to create an accurate representation for the surrounding environment and estimate the ego-state (i.e. self-localization) within it. The traditional methods for localization are on the basis of Global Positioning System (GPS) or Real-Time Kinematic (RTK) positioning system (Cadena et al., 2016b). However, the measurement error of GPS is limited to more than ten meters due to the signal reflections, clock errors and atmospheric conditions, it is unacceptable for vehicle navigation, especially when the vehicle travels in scenes of tunnels and urban canyons (Cheng et al., 2019). RTK is able to correct these errors through the inner correction signal of a fixed calibrated based station, but such system is reliant on additional infrastructure with high-costs (Infotip Service GmbH, 2019). Simultaneous Localization and Mapping (SLAM) method is considered to be a good solution for localization and navigation of autonomous driving vehicles, it can estimate the pose of the moving vehicles in real time with the map of the surrounding environment is constructed at the same time (Durrantwhyte and Bailey, 2006). According to the different types of the sensors, SLAM method can be mainly divided into two categories: Light Detection and Ranging (LIDAR) SLAM and visual SLAM. As the LIDAR SLAM is started earlier than visual SLAM, it is relatively mature in the applications of autopilot (Debeunne and Vivet, 2020a). Compared with camera, the LIDAR sensor is less sensitive to the changes of light and night. In addition, it can provide 3D map's information with a larger Field of View (FOV). However, the unaffordable cost and long development cycle on the large scale lead to the LIDAR sensors are difficult to popularize. In comparison, visual SLAM has the advantages of rich information and easy-to-install, as well it makes the system cheaper and lightweight. At present, visual SLAM system can run in micro Personal Computer(PC) and embedded device, even in mobile devices like smart phones (Klein and Murray, 2009).

Unlike the indoor or outdoor mobile robots in which majority of the developed visual SLAM methods are tested on, autonomous driving vehicles have more complex parameters should be taken into consideration, particularly when the vehicle driving autonomously in an urban environment. For example, the area of the environment is larger scale with dynamic obstacles thus making the performance of visual SLAM method is not accurate and robust enough (Cadena et al., 2016a). Issues such as error accumulation and illumination change and fast motion lead to problematic estimations. Various methods have been taken into consideration to solve these problems related to autonomous driving vehicles. Such as the feature point/direct/semi-direct/point-line fusion based algorithms for Visual Odometry (VO) (Singandhupe and La,

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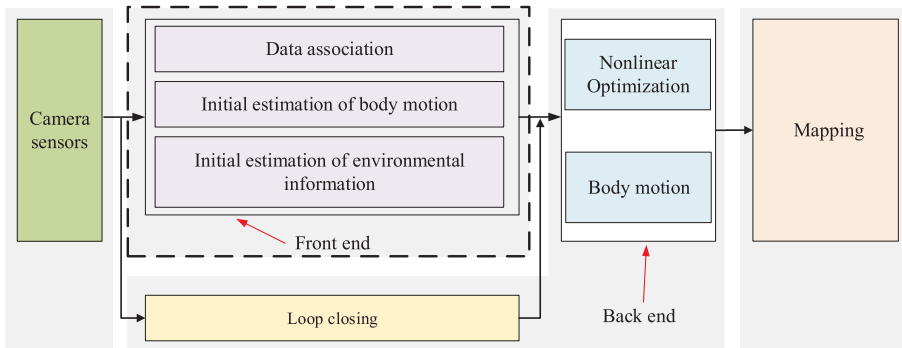


Fig. 1. The structure of a typical visual SLAM system.

2019), and Extended Kalman Filter (EKF)/graph-based optimization algorithms for pose estimation (Takleh et al., 2018). Also the visual based multi-sensor fusion methods have attracted great attention for improving the accuracy of autonomous system.

In terms of the visual based SLAM systems, except for the mapping module, the collection of sensors' data such as camera or Inertial Measurement Unit (IMU), VO and Visual-Inertial Odometry (VIO) systems are completed in the front end, and the optimization, loop closing is completed in back end. Re-localization is always considered to be an additional module for improving the accuracy of the visual SLAM system (Taketomi et al., 2017). In this paper, a review of the visual SLAM methods is presented. It is mainly considered from the aspect of positioning accuracy for visual SLAM systems, and the methods that might be applied to autonomous driving scenarios have been investigated in detail as much as possible, including pure visual SLAM methods, visual-inertial SLAM methods, and visual-LIDAR-inertial SLAM methods, as well the positioning accuracy of our previous work are compared with the well-known methods on the public datasets. This review provides a detailed survey of visual SLAM techniques and can serve as a friendly guide for new researchers in the field of autonomous driving vehicles. Also, it can be considered as a dictionary for experienced researchers to seek the possible directions in future works.

The remainder of this article is structured as follows: Firstly, we present an overview of the principle of the visual SLAM system, commenting on the responsibilities of the camera sensors, front-end, back-end, loop closing, and mapping modules in Section 2. Secondly, the state-of-the-art studies of visual, visual-inertial, visual-LIDAR, and visual-LIDAR-IMU SLAM methods are illustrated in Section 3. Thirdly, the key issues and the future development trend of visual SLAM technology for the application of autonomous driving vehicles are discussed in Section 4. Finally, the conclusion is drawn in Section 5.

2. Principles of visual SLAM

The classic structure of a visual SLAM system can be divided into five parts: Camera sensors module, front-end module, back-end module, loop closing module and mapping module. As shown in Fig. 1, the camera sensors module is responsible for collecting image data, the front-end module is responsible for tracking image features between two adjacent frames to achieve the initial camera motion estimation and local mapping, the back-end module is responsible for the numerical optimization from the front-end and further motion estimation, the loop closing module is responsible for eliminating accumulated errors by calculating image similarity in large scale environment, and the mapping module is responsible for reconstructing the surrounding environment (Gao et al., 2017).

2.1. Camera sensors

According to different types of sensors, the common visual sensors can be mainly divided into monocular, binocular, RGB-D and event cameras. The camera sensors are shown in Fig. 2.

(a). Monocular camera, the monocular camera has the advantages of low cost and simple layout. However, the depth of landmarks cannot be well-estimated, which suffers the scale ambiguous issue in the process of map building (Haomin et al., 2016). In addition, the pixel distance cannot be obtained when the camera is stationary (no translation) or only pure rotation.

(b). Binocular camera, by comparison with the monocular camera, the binocular SLAM can rely on the cameras' measurements to calculate the pixel depth in a static state, also it has been proved to be more robust than monocular SLAM in outdoor environment. However, the range of depth measurement is limited by the length of baseline and resolution, as well the process of parameter configuration and calibration is complex, since the camera needs to process the double image information, the calculation is more expensive for Central Processing Unit (CPU).

(c). RGB-D camera, different from monocular and binocular cameras, RGB-D cameras can directly obtain pixel depth through infrared structure-light and/or Time of Flight (TOF) which avoids complicated calculation. The theory of TOF is that the distance is calculated by measuring the time of laser's flight. However, due to the measurement range is narrow and it is extremely easy to be disturbed by sunlight, the RGB-D camera is considered to be not suitable for the outdoor application such as autonomous driving scenarios.

(d). Event camera, the event camera has a high dynamic range with 140 dB, while the other cameras are not up to 60 dB, besides it has high temporal resolution, low power consumption, and do not suffer from motion blur. The literature (Gallego et al., 2019) describes that instead of capturing images at a fixed rate, the event camera measures per-pixel brightness changes asynchronously. Therefore, the event camera has a better performance for ranging in high speed and high dynamic conditions. The event camera can be classified as dynamic visual sensor (Lichtsteiner et al., 2008; Son et al., 2017a; Posch et al., 2009; Hofstatter et al., 2010), dynamic line sensor (Posch et al., 2007), dynamic and active-pixel visual sensor (Brandli et al., 2014), and asynchronous time based image sensor (Posch et al., 2010).

The popular visual sensor manufactures and products on the market are listed as follows but not limited to:

- **MYNTAI**: S1030 Series (stereo camera with IMU), D1000 Series (depth camera), D1200 Series (apply to smart phones).
- **Stereolabs ZED**: Stereolabs ZED camera (Depth Range: 1.5 to 20 m).
- **Intel**: 200 Series, 300 Series, Module D400 Series, D415 (Active IR Stereo, Rolling shutter), D435 (Active IR Stereo, Global Shutter), D435i (Integrated IMU).

(a) Monocular (b) Binocular (c) RGB-D (d) Event

Fig. 2. Commonly camera sensors.

- **Microsoft**: Azure Kinect (Apply to microphone with IMU), Kinectc-v1(structured-light), Kinect-v2 (TOF).
- **Occipital Structure**: Structure Camera (apply to ipad).
- **Samsung**: Gen2 and Gen3 dynamic cameras and event-based visual solution (Son et al., 2017b).

Other depth cameras are listed as follows but not limited to **Pixpro SP1**,[1] **Leap Motion**,[2] **IMI**,[3] **Humanplus**,[4] **Orbbec Astra**,[5] **PrimeSense**,[6] **DUO**,[7] **Xtion**,[8] **Pico Zense**,[9] **Camboard**.[10] Other event cameras are listed as follows but not limited to **iniVation**,[11] **AIT**,[12] **SiliconEye**,[13] **Prophesee**,[14] **CelePixel**,[15] **Dilusense**.[16]

2.2. Front end

The front end for visual SLAM is known as Visual Odometry (VO). It is responsible for estimating the camera motion and feature orientation roughly based on the information of adjacent frames. In order to obtain the accurate pose with a fast response speed, an efficient VO is necessary. At present, the front end can be mainly divided into two categories: the feature based method and direct method (includes the semi-direct method) (Zou et al., 2020). In this section, we mainly review the feature based method for VO. About the semi-direct and direct methods such as SVO (Forster et al., 2017) and DSO (Engel et al., 2018) are described in Section 3.1.2.

The VO system based on the feature point runs more stably and is relatively insensitive to the light and dynamic objects. The feature extraction method with high scale and well rotation invariance can greatly improve the reliability and stability of the VO system (Chen et al., 2019). In 1999, Lowe (2004) proposed the Scale Invariant Feature Transform (**SIFT**) algorithm, which was improved and developed in 2004. The overall algorithm is divided into three steps to complete the extraction and description of image feature points. **(i)**. Construct a scale-space through the Gaussian difference pyramid approach and identify points of interest through the Gaussian differential function. **(ii)**. Determine the position and scale of each candidate's position, and then locate those key points. **(iii)**. Assigning pointing characteristics to key points to get a description child.

The SIFT feature performs non-deformation against rotation, scaling, and illumination changes, but it consumes a large amount of calculation. Speeded Up Robust Features (**SURF**) (Herbert et al., 2007) is an improvement of SIFT. It solves the shortcomings of a large amount of SIFT calculations and poor real-time performance as well as maintains the excellent performance of SIFT operator. Despite all this, the SURF algorithm has greater limitations when applied in real-time SLAM systems. On the basis of ensuring performance, a feature extraction algorithm that pays more attention to computing speed is proposed. In 2011, Viswanathan (2011) proposed a local corner detection method based on template and machine learning approach, i.e. Feature from Accelerated Segment Test (**FAST**) corner detection method. The FAST algorithm takes the pixel to be detected as the center of the circle, and when the grayscale difference between other pixels on the circle with a fixed radius and the pixel at the center of the circle is large enough, the point is considered to be a corner point. However, FAST corner points do not have direction and scale information, they do not have rotation and scale invariance. In 2012, Rublee et al. (2012) proposed the Oriented FAST and Rotated BRIEF (**ORB**) algorithm based on FAST corner points and BRIEF descriptors. The algorithm first builds an image pyramid on the image, then detects FAST key points and calculates the feature vector of a key point. The descriptor of ORB adopts the fast calculation speed of the binary string feature BRIEF descriptor (Michael et al., 2010), so the ORB calculation speed is faster than the FAST algorithm with real-time feature detection. Besides, the ORB is less affected by noise and has good rotation Invariance and scale invariance which can be applied to real-time SLAM systems. In 2016, Chien et al. (2016) compared and evaluated the SIFT, SURF and ORB feature extraction algorithms for VO application. Through a lot of tests on the KITTI dataset (Geiger et al., 2013), it can be concluded that the SIFT is the most accurate at extracting features, while the ORB is less computationally intensive. Hence, as an embedded computer with limited computing power, the ORB method is considered to be better suited for the applications of autonomous driving vehicles.

Other image feature descriptors for VO are listed as follows but not limited to **DAISY** (Tola et al., 2010), **ASIFT** (Morel and Yu, 2009), **MROGH** (Fan et al., 2011a), **HARRIS** (Wang et al., 2008), **LDA-Hash** (Fan et al., 2011b), **D-BRIEF** (Trzcinski and Lepetit, 2012), **Vlfeat** (Vedaldi and Fulkerson, 2010), **FREAK** (Alahi et al., 2012), **Shape Context** (Belongie et al., 2002), **PCA-SIFT** (Ke and Sukthankar, 2004).

2.3. Back end

The back end receives the camera pose estimated by the front end, and optimizes the initial pose to obtain a globally consistent motion trajectory and environment map (Sunderhauf and Protzel, 2012). Compared with the diversified algorithms of front end, the types of current back end algorithms can be mainly divided into two categories: Filter based method (such as Extended Kalman Filter (EKF) Bailey et al., 2006), and optimization based method (such as the factor graph Wrobel, 2001). They are described as follows.

(a). The filter based method, this method mainly uses the Bayesian principle to estimate the current state based on the previous state and the current observation data (Liu, 2019). The typical filter-based methods include Extended Kalman Filter (EKF) (Bailey et al., 2006), Unscented Kalman Filter (UKF) (Wan and Merwe, 2000) and Particle

[1] https://kodakpixpro.com/cameras/action-sport/sp1
[2] https://www.ultraleap.com
[3] http://www.imi-tech.com/home/
[4] http://www.human plustech.com/
[5] https://orbbec3d.com
[6] http://xtionprolive.com/primesense-carmine-1.09-3d-sensor-motion-camera
[7] https://duo3d.com
[8] http://www.xtion.com/
[9] https://www.pico-interactive.com/
[10] https://pmdtec.com/zh/
[11] https://www.inivation.cn/
[12] https://www.ait.ac.at/
[13] https://www.activesilicon.com/
[14] https://www.prophesee.ai/
[15] http://www.celepixel.com/
[16] http://www.dilusense.com/

Filter (PF) (Arnaud et al., 2000). Taking the typical EKF-based SLAM method as an example, it is relatively successful in the application of the small scale environment. However, due to the covariance matrix is stored, its storage capacity increases with the square of the state quantity, the application of large unknown scenes is always limited.

(b). The optimization based method, the core idea based on the non-linear optimization (graph optimization) method is to convert the back-end optimization algorithm into a form of graph, with the subject pose and environmental characteristics at different moments as vertices, and the constraint relationship between vertices is represented by edges (Liang et al., 2013). After the graph is built, the optimization-based algorithm is used to solve the subject's pose, so that the state to be optimized on the vertices better meets the constraints on the corresponding edges. After the optimization algorithm is executed, the corresponding graph is the subject movement trajectory and environmental map. At present, most mainstream visual SLAM systems use nonlinear optimization methods.

2.4. Loop closing

The task of loop closing is to allow the system to recognize the current scene based on sensor information and determine that the area has been visited when returning to the original place, which is to eliminate the accumulated error of the SLAM system (Newman and Ho, 2005). For visual SLAM, the traditional loop detection method mainly uses the Bag-of-Words (BoW) model (Galvez-LoPez and Tardos, 2012), the step of realization is: **(i)**. Constructing a word list containing K words through K-means clustering for the local features extracted from the image. **(ii)**. Expressing the image as a K-dimensional numerical vector according to the number of occurrences of each word. **(iii)**. Judging the difference of the scene, and identifying whether the autonomous driving vehicle has reached the identified scene.

2.5. Mapping

A basic component of an autonomous vehicle is the ability to build a map of the environment and locate it on the map. Mapping is one of the two tasks of the visual SLAM system (i.e. localization and mapping), which plays an important role in navigation, obstacle avoidance and environmental reconstruction for autonomous driving. Generally speaking, the representation of maps can be divided into two categories: metric map and topological map. The metric map describes the relative positional relationship between map elements, while the topological map emphasizes the connection relationship between map elements. For the classic SLAM system, the metric map can be further divided into sparse map and dense map, the sparse map contains only a small amount of information in the scene, which is suitable for positioning, and the dense map contains more information, which is beneficial for vehicles to perform navigation tasks according to the map.

3. State-of-the-art studies

In this section, the state-of-the-art studies for visual and visual-based (e.g. visual, visual-inertial, visual-LIDAR, visual-LIDAR-IMU) SLAM are surveyed. They are described as follows in detail.

3.1. Visual SLAM

Similar to the VO subsystem described in Section 2.2, according to the method of utilizing information from images, the pure visual SLAM system can be divided into two categories: feature-based and direct methods. The feature-based method refers to the estimation of the camera movement between the adjacent frames and the construction of an environmental map by extracting and matching the feature points. This method has the disadvantage of taking a long time to extract feature points and calculate descriptors. Therefore, some researchers suggest to discard the calculation of key points and descriptors, then the direct method comes into being (Zou et al., 2020). In addition, according to the type of sensors, the visual SLAM can be divided into monocular, binocular, RGB-D, and event camera-based methods. According to the density of map, it can be divided into sparse, dense and semi-dense SLAM, they are introduced as follows:

3.1.1. Feature-based method

In 2007, Davison et al. (2007) proposed the first real-time monocular visual SLAM system, **Mono-SLAM**. The results from real-time feature patch orientation estimation are shown in Fig. 3(a). The EKF algorithm is used in the back end to track the sparse feature points acquired from the front end, and uses camera pose and landmark point orientation as state quantity to update its mean and covariance. In the same year, Klein and Murray (2007) proposed a parallel tracking and mapping system, **PTAM**. It realized the parallelization of tracking and mapping work. The process of feature extracting and mapping is shown in Fig. 3(b),For the first time, it distinguished the front-end and the back-end and proposed a key-frames mechanism, through a non-linear optimization method. The key images are connected in series to optimize the motion trajectory and feature orientation. Many subsequent visual SLAM system designs have also adopted similar methods. In 2015, Mur-Artal et al. (2015) proposed **ORB-SLAM** which is a relatively complete key frame-based monocular SLAM method. Compared with the dual-thread mechanism of PTAM, this method divides the entire system into three threads: tracking, mapping, and loop closing. It should be noted that the process of feature extraction and matching (left column), map construction and loop detection are all based on ORB features (right column). Fig. 3(c) is the real-time feature extraction process (left column) and trajectory tracking and mapping result (right column) of the monocular camera in the college road environment. In 2017, The subsequent release of **ORB-SLAM2** (Murartal and Tardos, 2017) is proposed by Mur-Artal et al. it supports loop detection and relocation with the function of real-time map reuse, in addition the improved framework also opened the interface between the binocular camera and the RGB-D camera. The left column of Fig. 3(d) shows the binocular trajectory estimation and feature extraction of ORB-SLAM2. The right column of Fig. 3(d) shows the keyframe and dense point cloud mapping effect of an RGB-D camera in an indoor scene. The continuous small green squares in the picture constitute the trajectory of the keyframe, and the dense 3D scene map built by the RGB-D camera surrounds it.

3.1.2. Direct based method

In 2011, Newcombe et al. (2011b) proposed a monocular SLAM framework based on the direct method, **DTAM**. Different from the feature-based methods, DTAM adopts the inverse-depth based method to estimate the depth of the features. The pose of the camera is calculated by direct image matching, and a dense map (Fig. 4(a)) is constructed through the optimization-based method. In 2014, Jakob et al. (2014) proposed **LSD-SLAM** (Fig. 4(b)), which is a successful application of the direct method in the monocular visual SLAM framework. This method applies the pixel-oriented method to the semi-dense monocular SLAM system. Compared with the feature-based method, LSD-SLAM is less sensitive, but the system is fragile when the camera's internal parameters and illumination change. In 2017, Forster et al. (2017) proposed **SVO** (Semi-direct Visual Odometry). It uses the sparse direct method (also known as the semi-direct method) to track the key points (the bottom of Fig. 4(c)), and the pose is estimated based on the information around the key points. The trajectories with sparse maps in an indoor environment are shown in the top of Fig. 4(c). Due to the semi-direct method tracks sparse features, neither calculating descriptors nor processing dense information, SVO has low time complexity and strong real-time performance. In 2016, Engel et al. (2018) proposed **DSO**, which also uses the semi-direct method to ensure higher accuracy at a faster operation speed. However, as mentioned in Section 2.1, they

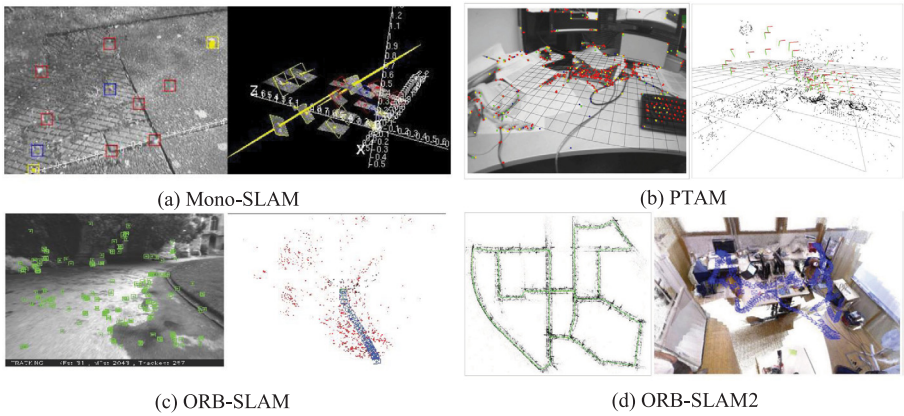


Fig. 3. (Davison et al., 2007; Klein and Murray, 2007; Mur-Artal et al., 2015; Murartal and Tardos, 2017) Typical feature based Visual-SLAM frameworks, (a) Mono-SLAM, real-time feature patch orientation estimation for an outdoor scene (b) PTAM, a desktop is tracked, the online constructed map contains about 3000 feature points. (c) ORB-SLAM, a period of time after initialization in the New-College sequence, (d) ORB-SLAM2, processing stereo (left column) and RGB-D (right column) inputs to estimate trajectory and construct the map.

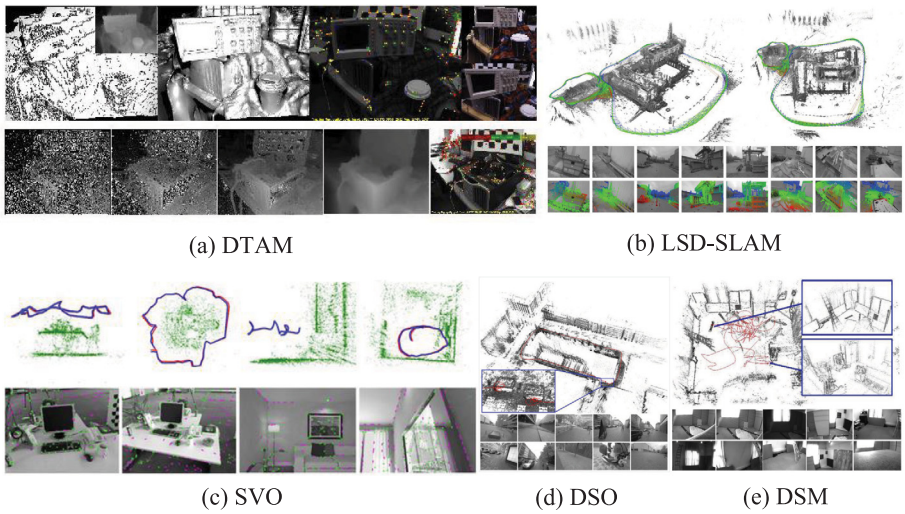


Fig. 4. (Newcombe et al., 2011b; Jakob et al., 2014; Zubizarreta et al., 2020; Forster et al., 2017; Engel et al., 2018) Typical direct Visual-SLAM frameworks. (a) DTAM, several dense maps are constructed through the optimization based method in an indoor scene. (b) LSD-SLAM, the sparse maps based on large outdoor scenes are reconstructed. (c) SVO, the estimated trajectories (blue color) is compared with ground truth (red color), it has been proved to have a good tracking performance in small scale scene. (d) DSO, The 3D reconstruction and tracking result (monocular visual odometry only). (e) DSM, qualitative example on V1_03_difficult sequence of EuRoC dataset, the trajectory is displayed in red color.

are just visual odometry. Due to the lack of a back-end optimization module and loop closing module, the tracking error of the system will accumulate with time. Fig. 4(d) shows the effect of the 3D reconstruction and tracking of the DSO (monocular visual odometer). The direct method has the advantages of fast calculation speed and insensitivity to weak feature conditions. However, it is based on the strong assumption that the gray level is unchanged, so it is very sensitive to changes in illumination. On the contrary, the feature point method has good invariant characteristics. In 2020, Zubizarreta et al. (2020) proposed a direct sparse mapping method, **DSM**, which is a full monocular visual SLAM system based on a photometric bundle adjustment (PBA) algorithm. The qualitative examples testing on the EuRoC dataset (V1_03_difficult sequence) are shown in Fig. 4(e). It has been proved to reduce both the estimated trajectory and map error while avoiding inconsistent map point duplications at the same time. The Table 1 summarizes the main characteristics with pros and cons of state-of-the-art visual SLAM frameworks. In addition to the aforementioned typical frameworks, there are other related works have been studied, such as **(i)**. Sparse visual SLAM: CubemapSLAM (Wang et al., 2018), ProSLAM (Schlegel et al., 2018), ENFT-SLAM (Zhang et al., 2016), OpenVSLAM (Sumikura et al., 2019), TagSLAM (Pfrommer and Daniilidis, 2019), UcoSLAM (Munoz-Salinas and Medina-Carnicer, 2019). **(ii)**. Semi-Dense visual SLAM: CNN-SVO (Loo et al., 2018), EVO (based on the event camera) (Rebecq et al., 2016). **(iii)**. Dense visual SLAM: MLM-SLAM (Nicholas Greene et al., 2016), Kinect Fusion (Newcombe et al., 2011a; Izadi et al., 2011), DVO (ucker et al., 2011; Kerl et al., 2013b,a), RGBD-SLAM (v2) (Endres et al., 2013), RTB-MAP (Labbe and Michaud, 2014; Labbäe and Michaud, 2011), Dynamic Fusion (Newcombe et al., 2015), Volume Deform (Innmann et al., 2016), Elastic Fusion (Whelan et al., 2015, 2016), Infini TAM (Prisacariu et al., 2017; Kahler et al., 2016), Bundle Fusion (Dai et al., 2017), KO-Fusion (Houseago et al., 2019), SOFT-SLAM (Cvisic et al., 2017). Other works can be listed as follows but not limited to RKD-SLAM (Liu et al., 2018b) and RGB-D SLAM (Dai et al., 2018). Maplab (Schneider et al., 2018a), PointNVSNet (Xu et al., 2019), MID-Fusion (Xu et al., 2018b) and MaskFusion (Runz et al., 2018). As you can see, there are a lot of achievements in the field of visual SLAM, we just provided a review of the popular approaches. Even if visual SLAM provides good positioning and mapping results, all of those solutions have merits and shortcomings. In this work, the advantages and disadvantages of "sparse-based method", "dense-based method", "feature-based method", "direct-based method", "Monocular method",

Table 1
Representative results of visual SLAM, – represents non-existent.

Benchmark	Frameworks	Back end	Sensor	Mapping	Pros	Cons	Year
Feature-based method	Mono-SLAM Davison et al. (2007)	Filter	Monocular	Sparse	With a persistent sparse map	Map updates are slow	2007
	PTAM Klein and Murray (2007)	Optimization	Monocular	Sparse	High reliability	Unknown scale	2007
	ORB-SLAM2 Murartal and Tardos (2017)	Optimization	Monocular	Sparse	High accuracy and support relocation	Scale uncertainty and poor initialization	2017
			Binocular	Sparse	Accurate depth measurement	Time-consuming	2017
			RGB-D	Dense	Good map reconstruction	Light sensitivity	2017
Direct-based method	DTAM Newcombe et al. (2011b)	Optimization	RGB-D	Dense	Good real-time performance	Easy to be affected by illumination changes	2011
	LSD-SLAM Jakob et al. (2014)	Optimization	Monocular	Semi-Dense	Low drift in large scale environment	Easy to be affected by illumination changes	2014
	DSM Zubizarreta et al. (2020)	Optimization	Monocular	Semi-Dense	With a persistent map	Large drift with geometric techniques alone	2020
Semi direct-based method	SVO Forster et al. (2017)	–	Monocular	Semi-Dense	Lightweight and good robustness	No loop closing	2014
	DSO Engel et al. (2018)	–	Monocular	Semi-Dense	Good robustness under weak texture	Not easy to maintain the global map	2018

Table 2
Summary of visual SLAM with advantages and disadvantages.

	Spare	Dense	Feature-based	Direct-based	Monocular	Binocular	RGB-D	Event camera
Advantages	Fast processing speed, light weight	Good map reconstruction	Relatively stable, not sensitive to motion and rotation	No feature detection needed, Semi-dense map reconstruction	Low cost sensor, the measurement distance is not limited	Depth can be measured directly, Suitable for indoor and outdoor	Depth can be measured directly, dense map reconstruction	Good adaptability under strong light, Infinite frame-rate
Disadvantages	easy to lose frames during rotation, especially for pure rotation, sensitive to noise	High occupancy and consumption	Sensitive to ambient light and texture, time consuming	Computational cost, required photometric calibration, often GPU based	Scale uncertainty, low initialization efficiency	Motion blur and insufficient dynamic range	work only indoor, very short range Sensitive to daylight	Sensors expensive, detect only changes of the environment

"Binocular method", "RGB-D method" and "event camera method" are summarized. It can be found in Table 2.

The determination of visual SLAM is technically challenging. Monocular visual SLAM has problems such as scale ambiguity, necessary initialization and scale drift (Strasdat et al., 2010). Although stereo and RGB-D cameras can solve the problems of scale and initialization, some obstacles cannot be ignored such as fast movement, large calculation, small field of view, occlusion, dynamic scenes, feature loss and changing light. Those issues limit the application of visual SLAM in the field of autonomous driving. At present, many researchers try to integrate different sensors into the VSLAM system for application. A typical multi-sensor fusion method is visual-inertial, visual-LIDAR (Yang, 2019; Ma et al., 2019), visual-LIDAR-IMU (Guilln et al., 2017; Zhang et al., 2018).

3.2. Visual-inertial SLAM

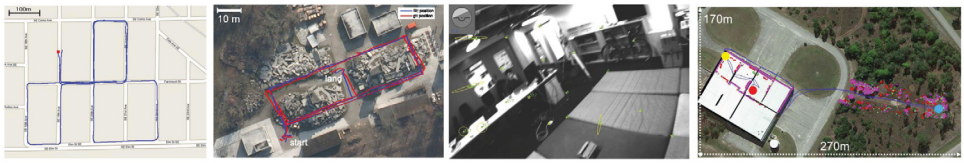
The IMU sensor can provide a good solution to for the problem of tracking failure when the camera moves into the challenging environment (less texture and/or lighting changes), on the other hand, the visual sensor can make up for the cumulative drift of the IMU. Such the combination of visual and IMU is known as golden partner. Due to the complementary function of camera and IMU have good development prospects in fields such as unmanned driving (Sun and Tian, 2019). The main methodology of VI-SLAM is to combine IMU information into the front-end of visual SLAM system, which is also known as Visual-Inertial Odometry (VIO) system. Generally, VI-SLAM systems can be divided into two categories: filter-based and optimization-based methods.

3.2.1. Feature-based method

In 2007, Mourikis and Roumeliotis (2007) proposed the Multi-State Constraint Kalman Filter (**MSCKF**), it is the earliest well known visual-inertial SLAM system based on the Extended Kalman Filter (EKF) algorithm. Compared with pure visual odometry, MSCKF (Fig. 5(a)) can adapt to more violent motion and texture loss for a certain period of time with higher robustness. In 2012, Stephan (2012) proposed SSF (Fig. 5(b)), it is a time delay compensated single and multi-sensor fusion framework based on an EKF with loosely-coupled method. In 2013, Li and Mourikis (2013) pointed out the inconsistency of MSCKF in the state estimation process. In 2017, Paul et al. (2017) proposed MSCKF2.0, which has greatly improved accuracy, consistency and computational efficiency. In addition, ROVIO (Robust Visual Inertial Odometry) (Bloesch et al., 2015) (Fig. 5(c)) and MSCKF-VIO (Ke et al.) (Fig. 5(d))are also excellent works based on filtering methods in recent years.

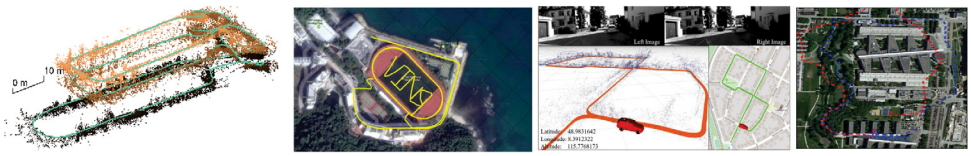
3.2.2. Optimization-based method

In terms of the optimization-based VI-SLAM systems, the most classic frameworks are **OKVIS** (Open Keyframe-based Visual-Inertial SLAM) (Leutenegger et al., 2015) (Fig. 6(a)) and **VINS-Mono,106** (Fig. 6(b)). In 2015, Leutenegge et al. proposed OKVIS which uses the IMU measurement values to predict the current state, spatial points and two-dimensional image features to constitute a re-projection error. The predicted IMU state quantity and optimized parameters constitute the IMU error term, then the re-projection error is combined with the IMU error term for optimization. In 2017, Tong et al. (2017) proposed VINS-Mono, it is regarded as an excellent monocular VI-SLAM system, which an optical flow method is used in the front-end, and a



(a) MSCKF (b) SSF (c) ROVIO (d) MSCKF-VIO

Fig. 5. (Mourikis and Roumeliotis, 2007; Stephan, 2012; Paul et al., 2017; Ke et al.) Feature based visual-inertial SLAM.



(a) OKVIS (b) VINS-MONO (c) VINS-Fusion (d) ORB-SLAM3

Fig. 6. (Leutenegger et al., 2015; Tong et al., 2017, 2019; Campos et al., 2020) State-of-the-art studies of visual-inertial SLAM.

nonlinear optimization algorithm based on sliding window is adopted in the back-end (Cheng et al., 2021b). In addition, the initialization method of VINS-Mono is worth to be noted, it adopts disjoint method (as well as **VI-ORBSLAM** Mur-Artal and Tards, 2017), which firstly initializes the pure visual subsystem, then estimates the bias, gravity, scale and speed of the IMU (accelerometers and gyroscopes). Through the tests of KITTI and EuRoC dataset, VINS-Mono has been proved to have a positioning accuracy equivalent to OKVIS with more complete and robust in initialization and loop closing stage. In 2019, the team of VINS-Mono proposed the binocular version as well the GPS information is integrated, VINS-Fusion (Tong et al., 2019). As shown in Fig. 6(c), due to the addition of GPS measurements, it has achieved good positioning and mapping effect in the outdoor environment, and it is also considered to be a good application in field of autonomous driving vehicles. In 2020, Campos et al. (2020) proposed a feature-based tightly-integrated visual-inertial SLAM system, **ORB-SLAM3**. It is the latest achievement with a more efficient initialization process by Maximum-a-Posteriori (MAP) algorithm, and it has achieved a multiple map function that relies on a new place recognition method with improved recall. Besides, the system is able to perform visual, visual-inertial and multi-map SLAM with monocular, binocular and RGB-D cameras. The experimental results of the outdoor scene are shown in Fig. 6(d). The pipeline of ORB-SLAM3 is similar to ORB-SLAM2, the whole system consists of three threads: Tracking, Local mapping and Loop closing (i.e. loop and map merging) threads. Besides, ORB-SLAM3 can survive long periods of poor visual information, when it gets lost, it starts a new map that will be seamlessly merged with previous maps when revisiting mapped areas. Table 3 summarizes the main algorithms in the visual-inertial SLAM frameworks in recent years.

At present, the Optimization-based VI-SLAM methods have become the mainstream. In addition to the above approaches, there are other state-of-the-art works can be summarized as follows but not limited to **BASALT** (Usenko et al., 2020), **Kimera** (Rosinol et al., 2020), **ICE-BA** (Liu et al., 2018a), **Maplab** (Schneider et al., 2018b), **StructVIO** (Zou et al., 2019). **RKSLAM** (Liu et al., 2016). The event camera based VI-SLAM system can be listed as follows but not limited to (Mueggler et al., 2018; Zhu et al., 2017; Nelson, 2019; Eckenhoff et al., 2019). The deep learning-based method can be seen in Jared Shamwell et al. (2019), it shows a network that performs VIO without IMU intrinsic parameters or the extrinsic calibration between an IMU and camera. The literature (Lee et al., 2019) provides a network to avoid the calibration between the camera and IMU sensors.

3.2.3. Testing and evaluation

In order to intuitively understand the positioning effect of the above SLAM method, some of the typical algorithms is tested and compared

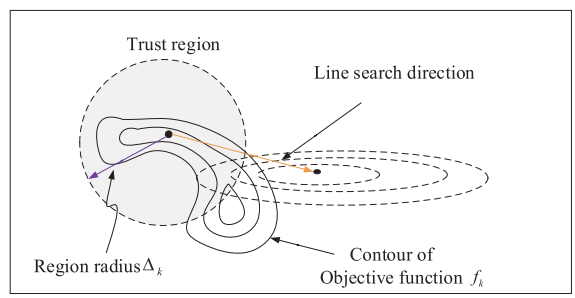


Fig. 7. The model of the trust-region based iterative strategy.

with one of our previous work (Cheng et al., 2021a) on the same on-board computer equipped with an Intel Core i7-9700 CPU, 16 GB RAM and Ubuntu18.04+Melodic operating system. As described in Cheng et al. (2021a), an improved trust-region iterative strategy was proposed based on the traditional Gauss–Newton (G-N) linear iterative strategy, then this strategy was integrated into VI-ORBSLAM framework (Mur-Artal and Tards, 2017) for faster initialization and higher positioning accuracy. The model of trust-region iterative strategy is shown in Fig. 7. It combines the steepest descent algorithm and G-N algorithm, the objective function is approximated with a trust model. While the solution is considered as the minimum of model function around the current point, then the minimization sub-problem is solved in each iteration step.

The initial parameters that need to be estimated include scale factor, velocity, gravity, and the biases of accelerometer and gyroscope. In order to make all variables observable, the pure ORB-SLAM system has been required to perform for a few seconds. The detailed steps of the method are described as follows: Firstly, the visual initialization process is performed, which includes ORB extraction, map initialization and initial pose estimation. Secondly, the IMU pre-integration technology is adopted for IMU-camera frequency alignment with the keyframes are generated. Thirdly, an improved iterative strategy based on the trust region is introduced for the gyroscope bias estimation as well as the gravity direction is refined. Finally, the accelerometer bias and visual scale are estimated on the basis of previous estimations. The pipeline of our previous work is shown in Fig. 8.

The 2D trajectories of algorithms on V2_01_easy sequence of EuRoC dataset are shown in Fig. 9. We can see that the test results of each algorithm deviate to varying degrees in comparison with the ground truth, the trajectory of our algorithm (red line) is closer to the ground truth (black dotted line), while the VI-ORBSLAM (blue line) suffers largest drift. The variation curves of position in X, Y, Z direction are

Table 3
Representative results of visual-inertial SLAM.

Benchmark	Frameworks	Coupled	Pixel	Sensor	Pros	Cons	Year
Filter-based method	MSCKF-VIO Mourikis and Roumeliotis (2007), Stephan (2012), Li and Mourikis (2013), Paul et al. (2017)	Loosely	Shi Tomasi	Monocular	Strong scalability	Poor accuracy	2017
				Binocular	Unlimited distance	Complex configuration	2017
	ROVIO Bloesch et al. (2015)	Tightly	Shi Tomasi	Monocular	High robustness with direct photometric feedback	Poor accuracy	2017
Optimization-based method	OKVIS Leutenegger et al. (2015)	Tightly	BRISK	Monocular	Lightweight	No loop closing and map	2014
				Binocular	Higher accuracy	Complex configuration	2014
	VI-ORBSLAM Mur-Artal and Tards (2017)	Tightly	ORB	Monocular	High accuracy with map reuse	Poor IMU initialization	2017
	VI-DSO Von Stumberg et al. (2018)	Tightly	High grad	Monocular	High accuracy and good robustness under weak texture	Not easy to maintain the global map	2018
	VINS-Mono Tong et al. (2017)	Tightly	Shi Tomasi	Monocular	Good real-time performance	Poor robustness during fast rotation	2018
	VINS-Fusion Tong et al. (2019)	Tightly	Shi Tomasi	Monocular	High accuracy with GPS integrated	Time consuming	2019
				Binocular	Low occupancy	Complex configuration	2019
				Fisheye	Wide field of view	High occupancy	2019
	BASALT Usenko et al. (2020)	Tightly	FAST	Binocular	High accuracy with globally consistent keyframe-based BA	Time consuming	2020
	Kimera Rosinol et al. (2020)	Tightly	Shi Tomasi	Binocular	Good real-time performance with 3D metric-Semantic mesh	Poor robustness	2020
	ORB-SLAM3 Campos et al. (2020)	Tightly	ORB	Binocular	long term navigation	Time consuming	2020
				Fisheye	Wide field of view	High occupancy	2020

shown in Fig. 10. And the comparison curves of Euler angle (i.e. Roll, Pitch, Yaw) are shown in Fig. 11. The quantitative Root Mean Square Error (RMSE) results and the frame rate in the same CPU platform (i7-9700 CPU) tested in the whole 11 sequences are shown in Table 4, since all algorithms adopt multithreading, the third column of Table 4 reports the frame rate when processing the image stream. As we can see, the VINS-MONO approach performances the maximum testing frame rate with the average value of 270.6fps, while the testing frame rate of ORB feature based approaches are 156.5fps (OURS), 133.7fps (ORB-SLAM3), 126.1fps (VI-ORBSLAM), respectively. The reason for this difference is that VINS-MONO does not extract features and descriptors in every image frame. In addition, the frame rate of our previous work is faster than ORB-SLAM3 and VI-ORBSLAM, it is because our method adopts an improved iterative optimization strategy in the initial stage. The RMSE and Cumulative distribution function (CDF) of the translation error are provided in Fig. 12 and Fig. 13, respectively, the RMSE and Cumulative distribution function (CDF) of the orientation error are provided in Fig. 14 and Fig. 15, respectively. In Fig. 12 and Fig. 14, the reported values are the median after 10 executions for each sequence and the best results are highlighted in bold font. It can be known that our previous work, a fast monocular visual-inertial system with an improved iterative strategy method for initialization, achieves the best accurate of positioning in almost all sequences. In practical, our method provides the best performance on six sequences (i.e. MH_01, MH_02, MH_03, MH_05, V1_03, V2_01) for the translation and seven sequences (i.e. MH_01, MH_02, MH_03, V1_02, V1_03, V2_01, V2_03) for the orientation, thanks to the excellent initialization process, even if the system fails to extract ORB features, the system can be restarted quickly for working. The VI-ORBSLAM has best performance in V1_01_easy sequence but could not complete run the V1_03_medium sequence (represented by the symbol x in Table 4), while the trajectories can be tracked completely by other algorithms. The failure for VI-ORBSLAM framework to run in V1_03_medium sequence is mainly because of the relatively harsh environment. In the scene of V1_03_medium sequence, the low texture and dark light condition leads to the loss of ORB feature tracking, while the system needs to be restarted, there is no rapid enough initialization process to make up for it. The latest research ORB-SLAM3 has the best performance in MH_04_difficult and V2_02_medium sequences, However, we found that the accuracy of ORB-SLAM3 framework in the test of almost every sequence is second only to our previous work, it is because the whole system (even in the IMU initialization stage) depends on the Maximum a Posteriori (MAP) estimation algorithm, which ensures the stable positioning performance in harsh environment. VINS-MONO has the best Orientation performance in MH_05_difficult sequence, different from other algorithms, VINS-MONO algorithm adopted the optical flow algorithm in the front end of system, it is relatively simple pixel-level tracking which is not easy to be lost while ORB descriptor template matching is much

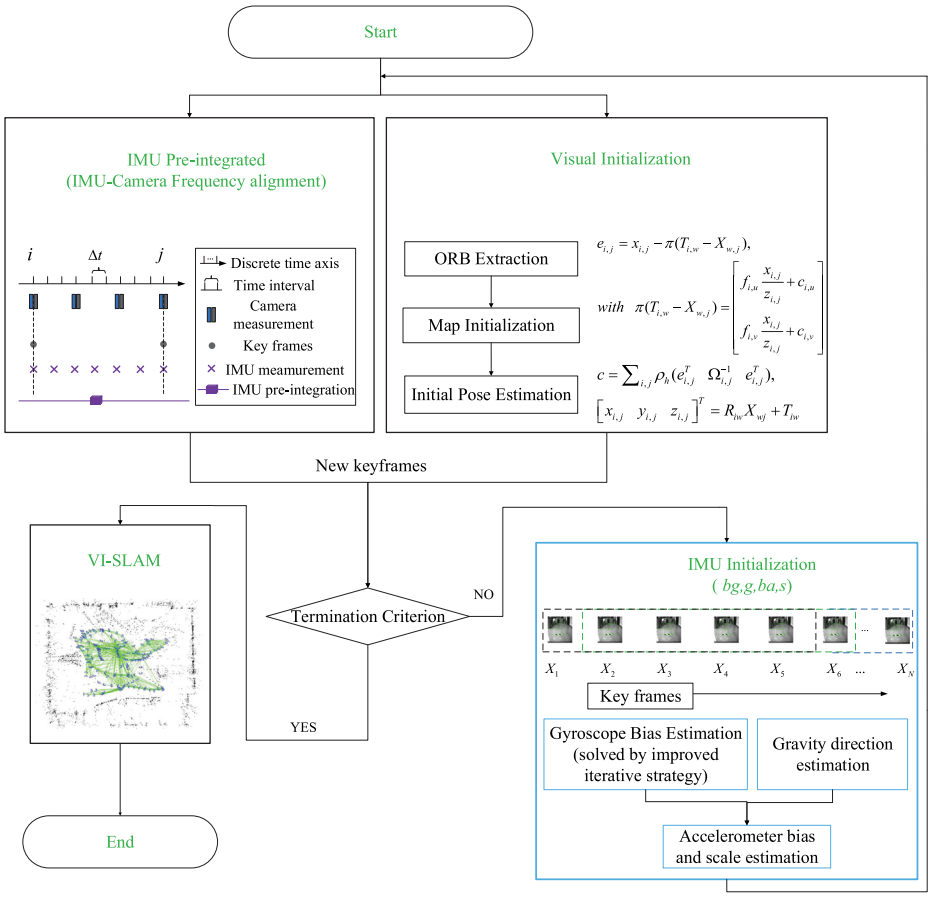


Fig. 8. The pipeline of our previous work. Abbreviations: bg—gyroscope bias, g—gravity, ba—accelerometer bias, s—visual scale, v—velocity.

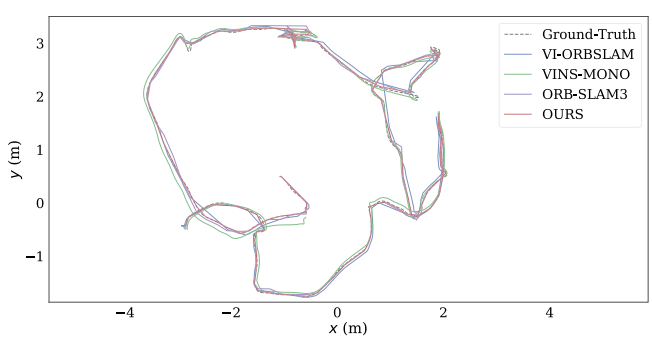


Fig. 9. 2D trajectories of state-of-the-art algorithms on V2_01_easy sequence.

stricter. In the scene of MH_05_difficult sequence, the ORB features are more likely to be lost compared with optical flow. In addition, the average values of the all sequences are calculated in the last row of the table, which VI-ORBSLAM is calculated as 0.077 m in translation and 1.971deg in orientation. VINS-MONO is calculated as 0.111 m in translation and 2.427deg in orientation. ORB-SLAM3 is calculated as 0.042 m in translation and 2.015deg in orientation. OURS is calculated with 0.036 m in translation and 1.909deg in orientation. Almost all of the state-of-the-art algorithms achieve high-precision positioning with centimeter level accuracy.

3.3. Visual-LIDAR SLAM

The visual and LIDAR have their own advantages, such as the visual can obtain a large amount of texture information from the environment, and has a strong scene recognition ability, while the LADAR not relying on light, well reliability, and higher accuracy in distance measurement. Therefore, in field of autonomous driving, the SLAM system that integrates visual and LIDAR can provide smarter and more reliable environmental perception and state estimation solutions. It follows a classical SLAM architecture with three main steps: **(i)**. The data processing step. **(ii)**. The estimation. **(iii)**. The global mapping step. According to the different proportions of visual and LIDAR in the SLAM system, the visual-LIDAR SLAM scheme can be divided into three categories: visual-led method, LIDAR-led method, and visual-LIDAR mutual correction method.

3.3.1. Visual-led method

The visual SLAM, especially for the monocular visual SLAM, is always inability to effectively extract the depth information of the feature points, while LIDAR is expert in it. In order to make up for the shortcomings of a visual SLAM, researchers tried to fuse LIDAR data into a visual SLAM system. The representative work of visual-led SLAM is **LIMO** (Graeter et al., 2018). This method projects the spatial point cloud which is obtained by the LIDAR on the image plane to estimate the scale of the visual features, then the visual feature scale recovered by the LIDAR and the feature scale construction error term derived from the camera pose estimation are treated as constraints for optimization in back-end. Shin et al. (2018b) proposed a method of using LIDAR to obtain sparse depth point cloud for visual SLAM, due to the resolution of the camera is much higher than that of LIDAR, this method suffers an issue that a large number of pixels without depth information. To deal with this problem, De Silva et al. (2018) adopted a Gaussian regression model to interpolate the missing depth value after calculating the geometric transformation between the two sensors. This method uses LIDAR to directly initialize the features detected in

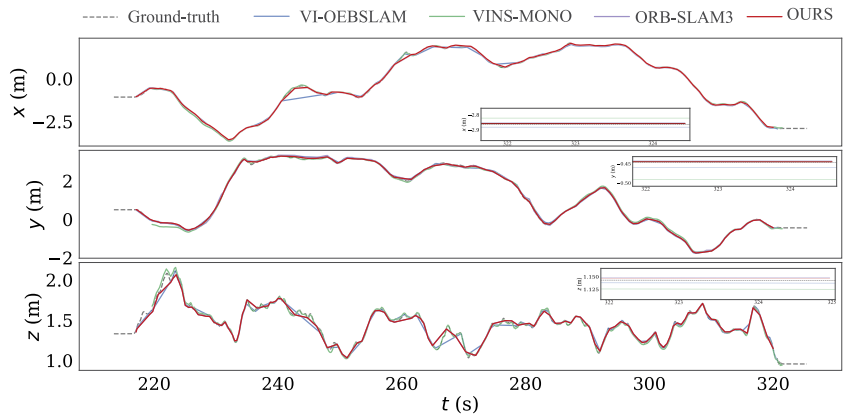


Fig. 10. Variation curves of positions in X, Y, Z direction.

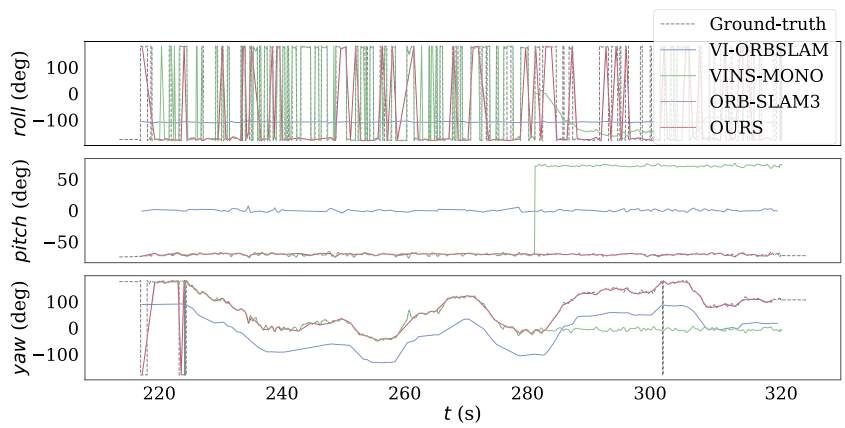


Fig. 11. Variation curves of Euler angles (i.e. Roll, Pitch, Yaw angles).

Table 4
The quantitative RMSE of the pose (translation and orientation) and Frame rate tested in the all sequences of EuRoC dataset. Abbreviations: Trans—Translation, Ori—Orientation, Fr—Frame rate, Ave—Average.

Seq.	VI-ORBSLAM			VINS-MONO			ORB-SLAM3(mono)			OURS		
	Trans. (m)	Ori. (o)	Fr. (fps)	Trans. (m)	Ori. (o)	Fr. (fps)	Trans. (m)	Ori. (o)	Fr. (fps)	Trans. (m)	Ori. (o)	Fr. (fps)
MH01	0.075	2.251	140	0.084	2.31	**275**	0.042	2.155	145	**0.031**	**2.036**	170
MH02	0.084	2.523	136	0.105	2.753	**275**	0.053	3.043	137	**0.015**	**2.408**	166
MH03	0.087	1.353	134	0.074	1.262	**274**	0.043	1.935	140	**0.026**	**1.156**	174
MH04	0.217	1.306	122	0.122	1.246	**272**	**0.099**	**1.106**	133	0.11	1.115	152
MH05	0.082	0.559	138	0.147	**0.395**	**266**	0.071	0.529	136	**0.056**	0.511	148
V101	**0.027**	**2.112**	123	0.047	2.733	**268**	0.043	2.422	140	0.031	2.657	153
V102	0.028	1.692	131	0.066	1.732	**269**	**0.016**	1.673	134	0.017	**1.646**	151
V103	x	x	119	0.18	4.598	**279**	0.025	2.632	122	**0.021**	**2.512**	159
V201	0.053	1.326	114	0.058	1.525	**263**	0.021	1.126	129	**0.013**	**1.069**	144
V202	0.041	2.58	119	0.09	2.613	**273**	**0.015**	**1.18**	131	0.03	2.394	159
V203	0.074	4.012	111	0.244	5.532	**263**	**0.037**	4.362	124	0.049	**3.496**	151
Ave.	0.077	1.971	126.1	0.111	2.427	**270.6**	0.042	2.015	133.7	**0.036**	**1.909**	156.5

the image, and its effect is the same as the method using the RGB-D sensor. There are also some studies that integrate LIDAR into visual SLAM to enhance the application value of the solution, such as reducing costs, improving performance, and enhancing system robustness. Some studies use the pose estimation of visual SLAM for point cloud annotation in the mapping stage. Zhang et al. (2018b) proposed a monocular visual SLAM method based on a single-dimensional LIDAR rangefinder, which has realized effective drift correction on low-cost hardware, and is used to solve the problem of scale drift that often occurs in monocular SLAM. Scherer et al. (2012) adopted drones to map the waterway and vegetation along the river, used a fusion framework combining visual odometry and IMU to estimate the state, and used LIDAR to detect obstacles and draw river boundaries, but this method produced the point cloud of contains occluded points, which reduces the accuracy of state estimation to a certain extent. Huang et al. (2019) solved this problem and proposed a direct SLAM method including occlusion point detection and coplanar point detection mechanisms.

3.3.2. LIDAR-led method

In terms of the LIDAR-led method, it adopts visual information to improve the accuracy of loop detection, or constructing a joint optimization function of LIDAR feature transformation error and visual re-projection error in the pose estimation stage to improve the robustness of pose estimation, such as Bai, etc. (Bai et al., 2016) used convolutional neural network to extract features to achieve loop detection, and effectively avoids mismatching of loop-closing scenes by setting the matching range, and ensures the real-time performance of the SLAM system through feature compression. Liang et al. (2016)

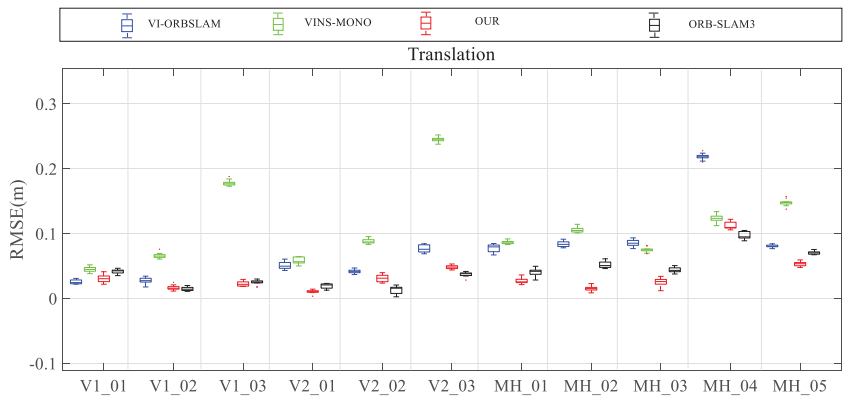


Fig. 12. RMSEs of translation. VI-ORBSLAM (blue color), VINS-MONO (green color), ORB-SLAM3 (black color), OURS (red color).

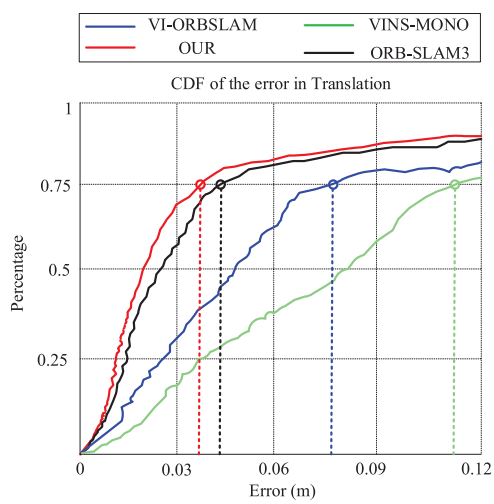


Fig. 13. Cumulative distribution function (CDF) of the translation error obtained by each approaches. VI-ORBSLAM (blue color), VINS-MONO (green color), ORB-SLAM3 (black color), OURS (red color).

used scan matching and ORB feature-based loop detection technology to improve the weak performance of LIDAR-based SLAM. Zhu et al. (2018) proposed a three-dimensional laser SLAM method using visual loop detection, which performs loop detection by using the key frame technology of visual bag-of-words. In addition, the Iterative Closest Point (ICP) method (Arun et al., 1987) can also be optimized by LIDAR and visual fusion. Pande et al. (2011) used visual information to estimate the rigid body transformation, and then proposed a generalized ICP framework.

3.3.3. Visual-LIDAR mutual correction method

The above research methods mostly use a single SLAM method, and use another sensor as an auxiliary device. There are also some studies that try to combine the two SLAM methods to correct each other. **V-LOAM** (i.e. Visual LIDAR Odometry And Mapping) (Zhang and Singh, 2015) is a classic real-time method of visual-LIDAR mutual correction. This method uses the camera pose estimated by the visual odometry within a circle of LIDAR scanning to correct the laser point. Cloud motion distortion, use the relative pose estimated from the LIDAR point cloud after adjacent scan correction to correct the visual estimated pose, and map the corrected point cloud to the local map for subsequent pose optimization. Seo and Chou (2019) proposed a parallel SLAM method that uses LIDAR SLAM and visual SLAM at the same time, which is characterized by using the measurement residuals of both modes at the back end to optimize the back end. Jiang et al. (2019) used both LIDAR constraints and feature point constraints to define the cost function of graph optimization, and constructed a 2.5D map to speed up the loop detection process. At present, the SLAM method based on visual-LIDAR fusion has less research results and practical applications than visual inertial fusion, and further exploration and research are needed.

3.4. Visual-LIDAR-IMU SLAM

At present, the multiple sensors fusion method (such as visual-LIDAR-IMU fusion SLAM) is considered to be applicable to L3 level's autonomous driving and has attracted the attention of many scholars. The SLAM system based on LIDAR can obtain a wide range of environmental details, but it is easy to fail in a scene (especially in the autonomous driving scene) lacking structural information. For example, long corridors or open squares. Vision-based method performs well in scenes with rich texture information, and it is easy to re-identify the scene (Shin et al., 2020). But it is very sensitive to changes in lighting, fast movements and initialization processes. Therefore, LIDAR and visual sensors are often fused with IMU to improve the accuracy and robustness of the system. IMU can remove the motion distortion of the point cloud, and last a short period of time in an environment lacking features, and at the same time can help the visual system to restore scale information (Geneva et al., 2019; Delmerico and Scaramuzza, 2018). At present, there are few research results on visual-LIDAR-IMU fusion SLAM (Debeunne and Vivet, 2020b). Some scholars try to use the visual-IMU fusion system (i.e. Visual Inertial System, **VIS**) and the LIDAR-IMU fusion system (i.e. LIDAR Inertial System, **LIS**) as the two separate modules are further fused to form a better performance visual-LIDAR-IMU fusion system (LIDAR Visual Inertial System, **LVIS**) (Chen et al., 2018). For the introduction of the visual-IMU fusion system, the corresponding description is made in Section 3.2, here we do not represent it repeatedly. In this review, the studies of laser-IMU fusion SLAM methods are also introduced as follows.

The schemes based on LIDAR-IMU fusion are divided into two categories: loosely coupled and tightly coupled schemes (Grant et al., 2013; Velas et al., 2016; Yang et al., 2018). Typical loose coupling schemes are **LOAM,146** (Fig. 16(a)) and **LeGO-LOMA** (Shan and Englot, 2018), in which IMU measurement information is not used in the optimization step. Compared with loosely coupled schemes, tightly-coupled schemes are in the development stage, which usually greatly improves the accuracy and robustness of the system. In the currently disclosed tightly coupled system, **LIO-Mapping** (Ye et al., 2019) uses the optimization process in VINS-Mono to minimize IMU residuals and LIDAR measurement errors. Since LIO-mapping is designed to optimize all measurement values, the real-time effect of the system is poor. Zou et al. proposed **LIC-Fusion** (Zuo et al., 2019), which is shown in the Fig. 16(b), it integrates both LiDAR features extracted in point clouds and sparse visual features, the LiDAR points in blue and red color are the plane and edge features, respectively. And the estimated

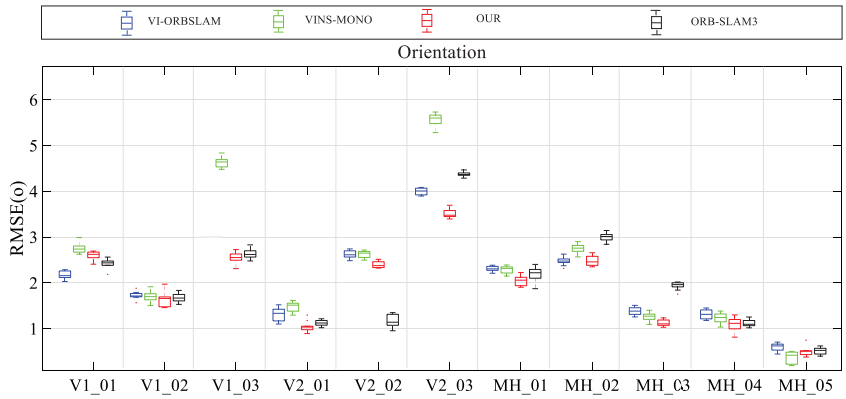


Fig. 14. RMSE of orientation. VI-ORBSLAM (blue color), VINS-MONO (green color), ORB-SLAM3 (black color), OURS (red color).

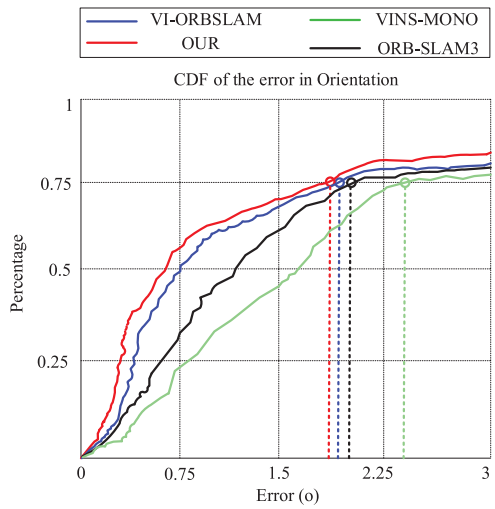


Fig. 15. Cumulative distribution function (CDF) of the orientation error obtained by each approaches. VI-ORBSLAM (blue color), VINS-MONO (green color), ORB-SLAM3 (black color), OURS (red color).

trajectory is marked in green color. Similar to LIO-mapping, it suffers the issue of poor real-time performance. In order to save computing resources, **LIO-SAM** (Shan et al., 2020) (Fig. 16(c)) introduced a sliding window optimization algorithm, using the factor graph method to jointly optimize the measurement constraints of IMU and LIDAR. **LINS** (Qin et al., 2020) (Fig. 16(e)), which is specially designed for ground vehicles, uses an error-state-based Kalman filter to iteratively correct the state quantity to be estimated. Zhang and Singh (2018) proposed a tightly coupled LVIO (LIDAR Visual Inertial Odometry) system, which uses a coarse-to-fine state estimation method, starting with IMU prediction for rough estimation, and then further refined by VIO and LIO. At present, the LVIO algorithm is the one with the highest test accuracy on the KITTI dataset. Zuo et al. (2019) realized the online calibration of spatiotemporal multi-sensor based on the MSCKF framework. Unfortunately, the code implemented by Zhang and Singh (2018) and Zuo et al. (2019) is currently not open source. Shan et al. (2021) released the latest visual-LIDAR-IMU tightly-coupled scheme in 2021: **LVI-SAM** (Fig. 16(d)). In order to improve the real-time performance of the system, it uses smoothing and mapping algorithms. The author regards visual-IMU and LIDAR-IMU as two independent subsystems. When enough feature points are detected, the two subsystems are linked together. When one of the subsystems fails to detect, the two can be separated independently for do not affecting each other. Table 5 summarizes the main algorithms in the visual-inertial SLAM frameworks in recent years.

Other solutions can be listed as follows but not limited to **VIL-SLAM** (Shao et al., 2019), **PointFusion** (Xu et al., 2018a), **Roar-Net** (Shin et al., 2018a), **AVOD** (Ku et al., 2018), **MV3D** (Chen et al., 2017), **FuseNet** (Hazirbas et al., 2007).

4. Discussions

Despite the great success of visual SLAM in localization and mapping of autonomous driving vehicles, as mentioned earlier, the existing technologies are not mature enough to fully solve the problem at hand. Current visual-based localization and mapping solutions are still in their infancy. To meet the autonomous driving requirements in complex urban environments, future researchers face many challenges. The practical application of these techniques should be considered as a systematic research problem. In addition, the SLAM system is only an integral part of the complex system of automatic driving vehicles, and the automatic driving system cannot completely rely on the SLAM system, it also needs to be equipped with modules such as control, target detection, path planning, and decision-making. In this section, we make some discussions about the overall observations and deductions with the current key issues of visual and visual based SLAM for the applications of autonomous driving vehicles, as well the future development trends. In this section, we make some discussions about the overall observations and deductions with the current key issues of visual and visual based SLAM for the applications of autonomous driving vehicles, as well the future development trends.

(a) Real-time performance: The application of autonomous driving vehicles requires the visual SLAM system responds as fast as possible. In the case of visual algorithms, the frequency of 10 Hz is considered to be the minimum desired frame rate for the vehicle to sustain autonomous driving on the urban roads (Bojarski et al., 2016). On the one hand, there are some visual algorithms have been proposed to explicitly optimize the real-time performance (Jaimez et al., 2017; Holzmann et al., 2016), on the other hand, they can be further improved through the hardware with a higher specifications performance such as Graphics Processing Units (GPUs). Besides, considering the accuracy and robustness of system, the various environmental dynamism (e.g. the scene changes, moving obstacles and illumination invariances (An et al., 2017; Kim and Kim, 2016; Liu et al., 2017)) should be taken into account. At present, the camera is most utilized to achieve obstacle detection or avoidance and lane-keeping for autonomous driving in specific scenarios, such as the Autonomous Valet Parking (APV) (Lovegrove et al., 2011).

(b) Positioning:The automatic driving of urban road scenarios is still at the technical tackling stage between L2 and L3, one of the key issues is that the position accuracy of vehicle is rough. We observed that the high-quality autonomous driving is inseparable from precision position in the order of centimeters for the vehicle to navigate even in unmapped environments. This precision is not achievable only depend

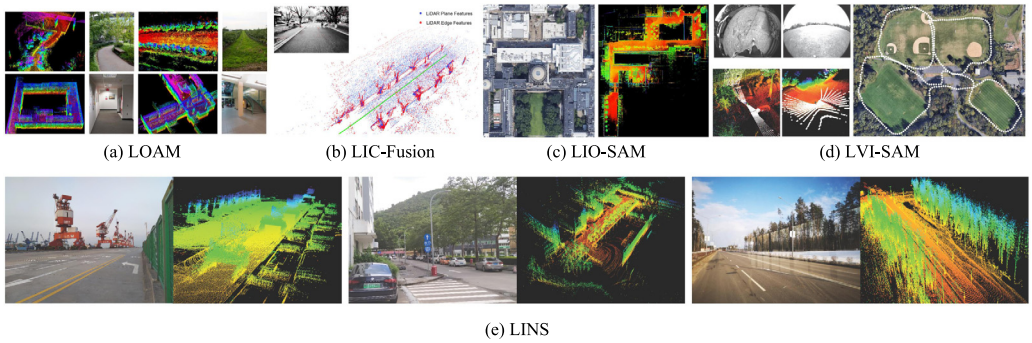


Fig. 16. (Zhang and Singh, 2017; Shan et al., 2020; Grant et al., 2013; Shin et al., 2020; Zuo et al., 2019) The results of visual-inertial-IMU SLAM(part), Almost all of the above applications are aimed at the scene of autonomous driving vehicles.

Table 5
Representative results of visual-LIDAR-IMU SLAM.

Benchmark	Frameworks	Coupled	Pros	Cons	Year
LIDAR-Camera fusion method	V-LOAM Zhang and Singh (2015)	Tightly	Low-drift, robust	Time consuming	2015
	LIMO Graeter et al. (2018)	Tightly	Accurate depth measurement	Poor performance for dynamic landmarks	2018
LIDAR-IMU fusion method	LOAM Zhang and Singh (2017)	Loosely	Low-drift and real-time	Difficult to perform loop closing and incorporate other absolute measurement	2017
	LeGO-LOAM Shan and Englot (2018)	Loosely	Low-drift and lightweight	High-cost, and less environmental information	2018
	LIC-Fusion Zuo et al. (2019)	Tightly	Online spatial and temporal calibration	Poor real-time performance and not open source	2019
	LIO-mapping Ye et al. (2019)	Tightly	Mapping with an efficient rotation-constrained	Poor real-time performance	2019
	LIO-SAM Shan et al. (2020)	Tightly	Low drift in large scale environment	Less environmental information without camera	2020
	LINS Qin et al. (2020)	Tightly	Lightweight and robust for ground vehicles in feature-less scenes	Computer consuming	
LIDAR-Camera-IMU fusion method	LVIO Zhang and Singh (2018)	Tightly	High robustness and low drift	Not open source	2018
	LVI-SAM Shan et al. (2021)	Tightly	Good real-time performance with loop detection	Poor robustness with fragile initialization	2021

on the conventional GPS receivers with accuracies of approximately 10 meters, expensive Differential GPS (DGPS) receivers are typically installed to achieve this, but it introduces redundancy while visual SLAM algorithm itself can be adopted for precise positioning. As reviewed in this paper, other GPS independent methods to achieve relative localization are studied, such as visual-inertial fusion method (Section 3.2), visual-LIDAR fusion method (Section 3.3), and visual-LIDAR-IMU fusion method (Section 3.4). In terms of the visual-inertial fusion method, unless the IMU is highly precise, the drift errors introduced by an IMU will exponentially affect accuracies. In terms of the visual-LIDAR fusion method, due to the lack of the self's Dead-Reckoning (DR) sensors (such as the encode and IMU sensors), the autonomous driving vehicle's robustness of localization is not guaranteed. In terms of the visual-LIDAR-IMU fusion method, to the best of our knowledge that there are no mature visual-based fusion SLAM algorithms applied to the real-world autonomous driving vehicle successfully, but there are many excellent fusion methods are being studied in latest years. With the reduction of the cost of LIDAR sensors, we believe that the visual-LIDAR-IMU fusion method is the final scheme of high-precision positioning for autonomous driving vehicles.

(c) Testing: At present, the real-world implementations are insufficiency, it can be credited to local legislation and the lack of development vehicles for autonomous drive tests. Here we observed that almost all of the recent visual SLAM works presented are tested on the public datasets (e.g. KITTI, EuROC, TUM etc.). Admittedly, they are great datasets for algorithm verification, but it remains to be verified how those algorithms will eventually perform in the real-world environment. Besides, testing on these datasets also limit the testing environments to the location where the dataset is captured, it might not an effective indicator for those algorithms in other countries or cities. Another reason for the lack of real-world implementation is seems that due to the higher computation requirement of visual SLAM algorithm, which indicates that an on-line implementation will require a computer with adequate and dedicated parallel processing hardware. The commonly mobile computers such as laptops do not possess the parallel computing capabilities of desktop GPUs. The commercial autonomous driving computers such as Nvidia's DRIVE PX2 (NVIDIA Corporation, 2017a) are so expensive that it is generally unaffordable for developments on a budget. It is inspiring to see that the recent availability of high performance and low-cost embedded devices with improved visual algorithms are emergence, such as the Nvidia Jetson (NVIDIA Corporation, 2017b) and the optimizations of fast VO (i.e. front end or the part of visual SLAM) methods (Jaimez et al., 2017; Jaimez and Gonzalez-Jimenez, 2015; Steinbrcker et al., 2011; Sun et al., 2018; Wu et al., 2017) could catalyze these implementations.

(d) Future development trends: Due to the complex modules of visual SLAM (e.g. front-end, back-end, loop closing and mapping, etc.), which increase the computational burden of the hardware platform, high-performance mobile computing platforms often limit the application of the above-mentioned visual SLAM algorithms in autonomous

driving. Multi-agent based visual SLAM technology seems to be able to overcome this problem. At present, multi-agent based visual SLAM is usually used in aerial drone, if it is installed on autonomous vehicles for mobile computing, the mobile computer platform is only responsible for processing the front-end data, while the process of back-end optimization and mapping are tackled by the remote server through the 5G/6G communication network transmit the data, we believe that it will greatly accelerate the application of visual SLAM for autonomous driving vehicles in the future.

5. Conclusion

The availability of recent researches have great contribution to the solution for the problem of visual SLAM. This work reviewed the various types of visual SLAM and/or visual based SLAM methods in relation to their applications for autonomous driving. At present, the application of visual SLAM in autonomous driving vehicle is considered to be not mature, but this still aroused widespread concern. Due to the easy attainability of the public datasets for autonomous driving, the visual SLAM algorithms are always easily to be verified, and the studies of new algorithms are encouraged. However, although the availability of datasets is promoting the proposal of new visual SLAM algorithms, a current shortage of real-world visual SLAM applications in the urban road environment still exist. In addition, the results from a dataset evaluation often deviates from a complete indication of the performance in a local real world environment, therefore a practical visual SLAM is expected to be emergence for autonomous driving vehicles' application. By reviewing state-of-the-art visual SLAM algorithms, it can be confirmed that the trend of current visual SLAM system is tending to be lightweight and multi-agent cooperation, it is encouraged to apply on the low-powered hardware such as embedded devices, and the multi sensor fusion algorithms are considered to be the core of visual SLAM application in autonomous driving vehicles. To sum up, there are still various issues in the field of autonomous driving, especially the combination of visual SLAM and autonomous driving vehicle require to be explored. Nevertheless, an increasing public recognition towards autonomous driving vehicles and forthcoming high-performance mobile computers will undoubtedly inspire the practical applications of visual SLAM in the near future.

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CRediT authorship contribution statement

Jun Cheng: Preparation, creation and/or presentation of the published work, Writing – original draft, Substantive translation. **Liyan Zhang:** Ideas, Formulation or evolution of overarching research goals and aims. **Qihong Chen:** Preparation, creation and/or presentation of the published work by those from the original research group, specifically critical review, commentary or revision – including pre- or postpublication stages. **Xinrong Hu:** Oversight and leadership responsibility for the research activity planning and execution, including mentorship external to the core team. **Jingcao Cai:** Verification, whether as a part of the activity or separate, of the overall replication/reproducibility of results/experiments and other research outputs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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